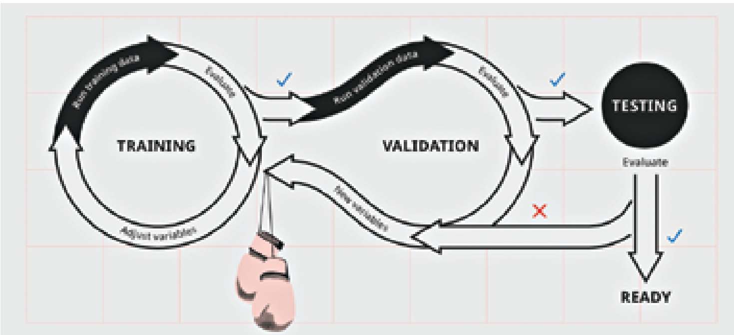
- **Validation dataset** is a subset of the training data. It is used to tune the hyperparameters (architecture) of the model and prevent overfitting. It checks how well the algorithm is performing during training so we can decide when to stop. If the accuracy over the training data set increases, but the accuracy over the validation data set stays the same or decreases, then the model is overfitting and should stop training.
- **Testing dataset** is kept hidden from the model during the training process. It is passed to the model after training to give an estimate of the unbiased predictive power of the model.



The process of training a model.

Training Dataset

Machine learning models are similar to how human children learn. Children depend on constant feedback from the environment and the object it is interacting with to learn. Machines learn when they see enough relevant data. This relevant data used to teach machines is known as the training data. Using this you can model algorithms to detect patterns, understand complex problems, find relationships, and make decisions. The quantity, quality and variety and of your training data will determine the success of your machine learning model. This dataset is usually prepared either by humans or by collecting data in a semi-automated way.

Why Training Data matters?

High quality and quantity of training data increase your model accuracy. In a real-world application, you should constantly provide data to continue improving its performance. But finding such reliable data at a large scale to train an accurate model is usually not possible.